

Introduction to Robotics

Robotics is an interdisciplinary field that combines mechanical engineering, electrical engineering, and computer science to design, construct, and operate robots capable of performing tasks autonomously or semi-autonomously. A robot typically consists of several key components, including a mechanical structure that provides physical form and movement capability, sensors that allow the robot to perceive its environment, actuators such as motors that enable movement and physical interaction with the world, and a control system that processes sensor information and determines appropriate actions. Robots can be broadly categorized based on their application, including industrial robots used in manufacturing settings for tasks such as welding, assembly, and painting, service robots designed to assist humans in settings such as healthcare or hospitality, and mobile robots capable of navigating and moving through various environments. The field of robotics draws heavily on principles from kinematics, which describes the motion of robotic systems without considering the forces that cause the motion, and dynamics, which accounts for the forces and torques involved in producing that motion. Control theory plays a central role in robotics, providing the mathematical framework necessary to design systems that can accurately follow desired trajectories or maintain stability despite external disturbances. As artificial intelligence and machine learning techniques have advanced, modern robots increasingly incorporate these technologies to improve their ability to perceive complex environments, make autonomous decisions, and adapt to changing conditions. The applications of robotics continue to expand rapidly, spanning manufacturing, healthcare, agriculture, space exploration, and household assistance, reflecting the growing role robots play in both industrial and everyday human life.